

SOLVING FOR INITIAL PRESSURE GRADIENT WITH SMOOTH INTERFACES

It is necessary to demonstrate pure hydrostatic equilibrium in the initial conditions of a two-fluid planar Rayleigh-Taylor instability. Any spurious wave phenomena can inject artificial energy into the mixing layer.

For initialization, pressure is estimated in a mass-weighted mixture-averaged sense m using the ideal gas law $P = \rho_m R_m T_m$. Here temperature is a constant for the fluids $T_m = T$. For the heavy h and light ℓ fluids, $\rho_m = \rho_h Y_h + \rho_\ell Y_\ell$. Therefore, from the law of partial pressures,

$$R_m = \frac{\hat{R}}{\frac{Y_h}{R_h} + \frac{Y_\ell}{R_\ell}}.$$

where $\hat{R}$ is the universal gas constant and the species gas constants are related through their molecular weights, $R_h \equiv \hat{R}/M_h$, $R_\ell \equiv \hat{R}/M_\ell$.

The initial free surface is perturbed by pseudo-random noise $\eta(x, y) \sim \sum \sin(x) \sin(y)$ in an (x, y) plane. The heavy (top) fluid mass fraction Y_h in the vertical z plane can then be approximated with a smooth interface using the hyperbolic tangent function as the level set function. It is of second-order accuracy and satisfactory of mass conservation in the region bounded by the interface. At the interface,

$$Y_h = \frac{1}{2} \left(1 + \tanh \left(\frac{z - \eta(x, y)}{\Delta} \right) \right)$$

where Δ is some parameter to control the thickness of smoothing. Here it is chosen in reference to the size of the domain $\Delta = C_\Delta \cdot 2\pi/N_p$, where N_p is the number of grid points in the z dimension and $C_\Delta = 4$. Enforcing mass conservation $\sum_k Y_k = 1$, $Y_\ell = 1 - Y_h$.

Specifying the initial 1-D vertical pressure gradient as only a function of gravity (body forces),

$$\frac{\partial P}{\partial z} = \rho_m g = \frac{P}{R_m T} g = \frac{P g}{\hat{R} T} / \left[\frac{\frac{1}{2} \left(1 + \tanh \left(\frac{z - \eta(x, y)}{\Delta} \right) \right)}{R_h} + \frac{1 - \frac{1}{2} \left(1 + \tanh \left(\frac{z - \eta(x, y)}{\Delta} \right) \right)}{R_\ell} \right].$$

Since η is independent of z this is a separable first-order linear ODE.

$$\frac{dP}{P} = \frac{g dz}{\hat{R} T} / \left[\frac{\frac{1}{2} \left(1 + \tanh \left(\frac{z - \eta(x, y)}{\Delta} \right) \right)}{R_h} + \frac{1 - \frac{1}{2} \left(1 + \tanh \left(\frac{z - \eta(x, y)}{\Delta} \right) \right)}{R_\ell} \right]$$

Then integrating and rearranging the fraction ($\frac{1}{1/\phi} = \phi$),

$$\begin{aligned} \int \frac{dP}{P} &= \ln(|P|) + C = \frac{g}{T} \int \left[\frac{\frac{1}{2} \left(1 + \tanh \left(\frac{z - \eta(x, y)}{\Delta} \right) \right)}{R_h} + \frac{1 - \frac{1}{2} \left(1 + \tanh \left(\frac{z - \eta(x, y)}{\Delta} \right) \right)}{R_\ell} \right] dz \\ &= \frac{g}{\hat{R} T} \left[\frac{z}{2R_h} + \frac{z}{2R_\ell} + \frac{\int \tanh \left(\frac{z - \eta(x, y)}{\Delta} \right) dz}{2R_h} - \frac{\int \tanh \left(\frac{z - \eta(x, y)}{\Delta} \right) dz}{2R_\ell} \right] \\ &= \frac{g}{\hat{R} T} \left[\frac{z}{2R_h} + \frac{z}{2R_\ell} + \frac{\Delta \cdot \ln \left(\cosh \left(\frac{z - \eta(x, y)}{\Delta} \right) \right)}{2R_h} - \frac{\Delta \cdot \ln \left(\cosh \left(\frac{z - \eta(x, y)}{\Delta} \right) \right)}{2R_\ell} \right] \\ &= \frac{g \left[z(R_\ell + R_h) - \Delta \cdot R_h \ln \left(\cosh \left(\frac{z - \eta(x, y)}{\Delta} \right) \right) + \Delta \cdot R_\ell \ln \left(\cosh \left(\frac{z - \eta(x, y)}{\Delta} \right) \right) \right]}{2\hat{R} T R_\ell R_h} \end{aligned}$$

Note that $\int dP/P = \ln(|P|) + C$, and $\int \tanh(z) dz = \ln(\cosh(z)) + C$. Lastly rearranging to obtain a common denominator $2R_h R_\ell$, the closed form solution can be written as,

$$P(z) = C \exp \left(\frac{g \left[z(R_\ell + R_h) - \Delta \cdot R_h \ln \left(\cosh \left(\frac{z - \eta(x, y)}{\Delta} \right) \right) + \Delta \cdot R_\ell \ln \left(\cosh \left(\frac{z - \eta(x, y)}{\Delta} \right) \right) \right]}{2\hat{R} T R_\ell R_h} \right).$$

With considerable algebra, this can be rewritten in the form $P(z) = C \exp(\mathcal{A}z) \cosh^{\mathcal{B}}\left(\frac{z-\eta}{\Delta}\right)$, where

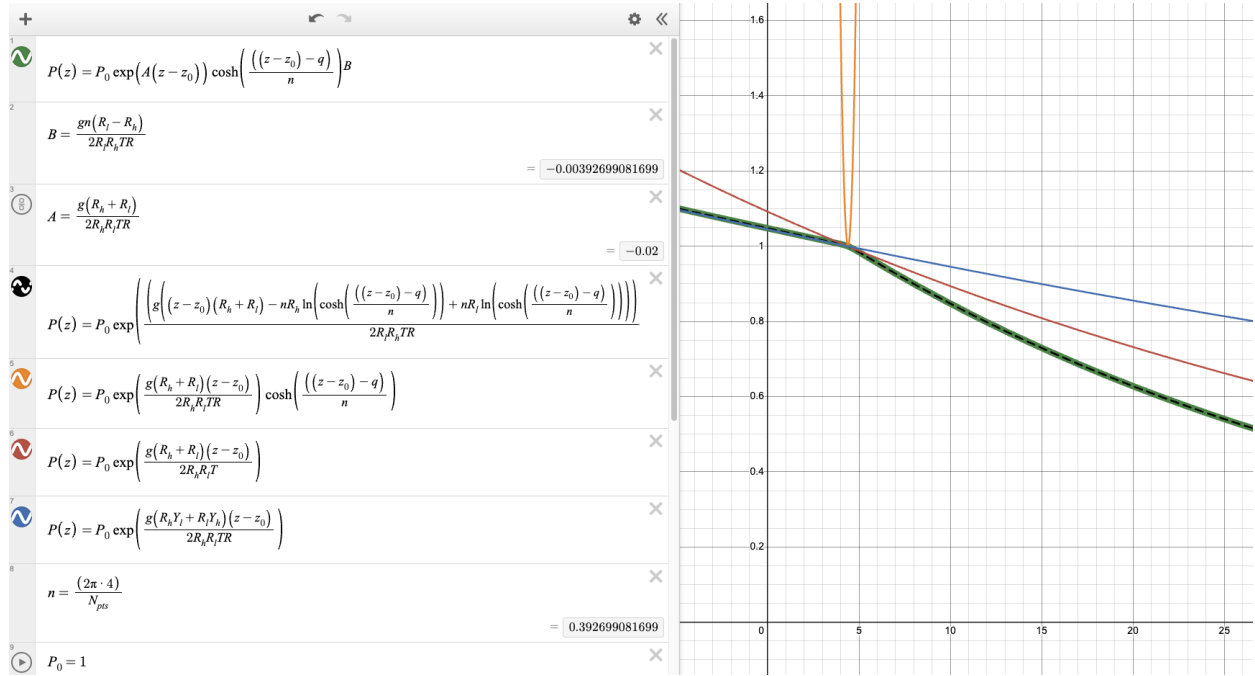
$$\mathcal{A} \equiv \frac{g(R_h + R_\ell)}{2\hat{R}TR_\ell R_h}, \quad \mathcal{B} \equiv \frac{g\Delta(R_\ell - R_h)}{2\hat{R}TR_\ell R_h}$$

Note that for $Y_h, Y_\ell = \text{constant}$ (no smoothing), the solution is

$$P(z) = C \exp\left(\frac{g(R_h Y_\ell + R_\ell Y_h)z}{2\hat{R}TR_h R_\ell}\right).$$

For a single material the solution takes the form $P(z) = C \exp(\frac{g}{R_1 T} z)$, where $R_1 = \hat{R}/M_1$ is the gas constant for a single material. In the limit $g/RT \gg L$, where L is the domain height, it can be seen that capturing the pressure distribution, especially in a mixing layer, will yield large truncation errors unless using a high-order scheme.

In order to initialize the instability at the center of the domain, z should be shifted by some offset $z - z_0$. For $z = z_0$ being the initial location of the free surface of the fluid interface, the initial condition is $P(z = z_0) = P_0$ such that $C = P_0$.



Some comments on the behavior:

- **cosh** is much “stronger” than **exp** and the exponent $\mathcal{B}$ serves to weaken it.
- In comparison with no smoothing, the **cosh** term still increases $P(z)$ considerably.

Based on simple intuition of the hydrostatic equation, initialized with heavy fluid on the right (steeper slope $P = \rho_h g z$) and lighter fluid on the left (shallow slope $P = \rho_\ell g z$). For the offset $P_0(z_0 = 1.4\pi) = 1$,

$$\text{if}(z \leq z_0) \rightarrow P = P_0 + \rho_h g(z - z_0)$$

$$\text{if}(z \geq z_0) \rightarrow P = P_0 - \rho_\ell g(z - z_0)$$

The pressure profile should look as follows:

